3.6 A Bound on Slopes

WORKSHEET

The question

The function f is defined by

$$f(x) = ax^2 + bx + c.$$

Show that

$$f'(x) = f(1)\left(x + \tfrac{1}{2}\right) + f(-1)\left(x - \tfrac{1}{2}\right) - 2f(0)x.$$

If a, b and c are real and such that $|f(x)| \leq 1$ for $|x| \leq 1$, show that $|f'(x)| \leq 4$ for $|x| \leq 1$.

Find particular values of a, b and c such that, for the corresponding function f of the above form, $|f(x)| \leq 1$ for all x with $|x| \leq 1$ and $|f'(x)| = 4$ for some x satisfying $|x| \leq 1$.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself The formula for $f'(x)$ uses only three values of f: at -1, 0 and 1. Why should three values be enough to determine the derivative of a quadratic?

STAGE 2 The bound

Ask yourself Take absolute values in the formula and use the triangle inequality, together with $|f(1)|, |f(-1)|, |f(0)| \leq 1$. What bound on $|f'(x)|$ do you get, as a function of x?

Pause and try Show that $\left|x+\frac{1}{2}\right| + \left|x-\frac{1}{2}\right| + 2|x| \leq 4$ for $|x| \leq 1$. (Split into cases according to the signs.)

STAGE 3 An extreme example

Ask yourself For the bound to be attained at $x = 1$, every inequality in Stage 2 must be an equality. What values must $f(1)$, $f(-1)$ and $f(0)$ take, and with what signs?

Try this next

1 **The general idea.** Show that if f is a quadratic then $f(x) = f(-1)\dfrac{x(x-1)}{2} + f(0)(1-x^2) + f(1)\dfrac{x(x+1)}{2}$, and differentiate to recover the formula in the question.

2 **Uniqueness.** HARDER Show that $\pm(2x^2-1)$ are the only quadratics with $|f(x)| \leq 1$ on $[-1,1]$ and $|f'(x)| = 4$ somewhere in $[-1,1]$.

3 **Chebyshev polynomials.** Use $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ to check that $T_3(x) = 4x^3 - 3x$, and verify that $T_3'(1) = 9$. Show that $T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$, and find T_4.

4 **Slopes of T_n.** By differentiating $T_n(\cos\theta) = \cos n\theta$ with respect to θ, show that $T_n'(\cos\theta) = \dfrac{n\sin n\theta}{\sin\theta}$, and deduce that $T_n'(1) = n^2$ by letting $\theta \to 0$.

Further reading

- **Lloyd N. Trefethen, *Approximation Theory and Approximation Practice*** (SIAM). A beautiful modern book about Chebyshev polynomials and interpolation, readable by a strong school student, with computer experiments throughout.
- **Peter Borwein & Tamás Erdélyi, *Polynomials and Polynomial Inequalities*** (Springer). For the full story of Markov's inequality and its relatives. Advanced.
- **George M. Phillips, *Interpolation and Approximation by Polynomials*** (Springer). An undergraduate text covering Lagrange interpolation, Chebyshev polynomials and Markov-type inequalities.

The full guided solution, and what the question leads to at university, is in Section 3.6:
https://beyondstep.keithmcnulty.org/ch03/markov-bound.html